3E3 Modelling Risk Coursework

Integrated Risk Analysis for GreenCo Energy

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1 Introduction

GreenCo Energy is developing a coastal renewable energy hub integrating offshore wind generation, on-site hydrogen production via electrolysis, and a heavy goods vehicle (HGV) refuelling station. The project faces four interlinked sources of uncertainty: **demand volatility** (stochastic monthly hydrogen demand), **production variability** (electrolyser availability), **congestion risk** (queueing at the refuelling station), and **long-term investment uncertainty** (demand growth trajectory and competitor entry).

This report develops an integrated risk analysis. Four models are applied, each addressing a different risk source: (1) *Newsvendor model* for the monthly production decision under demand uncertainty; (2) *M/M/s queueing model* for refuelling station congestion; (3) *Decision tree* for multi-stage strategy evaluation under scenario uncertainty; (4) *Monte Carlo simulation* for full NPV distributions and downside risk measurement. The objective is to maximise expected NPV while explicitly assessing downside risk exposure and service-level compliance.

2 Problem Setup

2.1 Analytical Models

We use the newsvendor inventory production model for computing the expected profit $\mathbb{E}[\Pi_t]$ and optimal production quantity at year t, Q_t^*. We also use a M/M/s queueing model to model refuelling service. Our parameters for these models, and demand growth models, are shown in Table 1. Exact details are explained in section 3.

Table 1: Model parameters.

Symbol	Parameter	Value
Demand & Pricing		
μ_0	Initial mean monthly demand	1000 t
σ_0	Initial demand std. deviation	150 t
CV	Coefficient of variation (σ_t/μ_t)	0.15
μ_t	Year-t mean monthly demand	Eqn (1)
σ_t	Year-t standard deviation of monthly demand	Eqn (1)
p	Selling price	£4000/t
c_p	Production cost	£2500/t
c_s	Spot market purchase price	£3200/t
c_h	Storage/handling cost (unsold)	£300/t
Production Capacity		
K_{nom}	Nominal electrolyser capacity	40 t/day
α	Operational availability	85%
n_d	Days per month	30
Refuelling Station		
λ	HGV arrival rate	10 vehicles/hr
$1/\mu$	Mean refuelling time	12 min
s_0	Initial number of bays	3
$W_q^{\max}$	Maximum average queue wait	5 min
Growth & Discounting		
p_H	Probability of high growth	0.45
g_H, g_L	Annual growth rates (mean demand)	10%, 3%
p_C	Probability of competitor entry (high growth, Y3)	0.40
δ	Demand reduction from competitor	15%
r	Discount rate	7%/yr

2.2 Demand Growth and Scenarios

Monthly hydrogen demand in year t: $D_t \sim \mathcal{N}(\mu_t, \sigma_t^2)$, with constant $\mathrm{CV} = \sigma_t/\mu_t = 0.15$, and annual growth rate of mean demand g ($g_H = 0.1$ for high growth, and $g_L = 0.03$ for low growth) gives:

$$\mu_t = \mu_0(1+g)^t, \qquad \sigma_t = \mathrm{CV} \cdot \mu_t \tag{1}$$

With competitor entry (high growth scenario, $t \geq 3$): $\mu_t^{\text{comp}} = (1-\delta)\mu_t$. This gives three terminal scenarios:

Table 2: Scenario probabilities and monthly demand (μ_t in t/month, σ_t in t/month).

Scenario $\mathbb{P}$(scenario)	**High, no comp.** $p_H(1-p_C) = 0.27$		**High + comp.** $p_H p_C = 0.18$		**Low growth** $1-p_H = 0.55$	
Yr	μ_t	σ_t	μ_t	σ_t	μ_t	σ_t
0	1000	150	1000	150	1000	150
1	1100	165	1100	165	1030	155
2	1210	182	1210	182	1061	159
3	1331	200	1131	170	1093	164
4	1464	220	1245	187	1126	169

2.3 Investment Strategies

GreenCo must choose one of three strategies at Year 0 to scale its hydrogen infrastructure over a 4-year horizon. Each strategy modifies a different subset of the newsvendor parameters $(c_s, \sigma_t, \bar{Q})$ and the queueing parameter s, creating distinct risk–return profiles.

Strategy 1 (Immediate Scale-Up). Invest £54M upfront (£48M electrolyser expansion + £6M for two additional refuelling bays) to increase capacity from $K_{\text{nom}} = 40$ to 65 t/day at 95% availability, giving $\bar{Q}_{S1} = 1852.5$ t/mo, and $s = 5$ bays. No changes to demand uncertainty or spot pricing. The bet is that high capacity eliminates spot procurement: since $\bar{Q}_{S1} \gg \mu_t$ for all t, the capacity constraint in the newsvendor never binds. However, the risk is overcapacity under the low growth scenario — demand utilisation $\mu_t/\bar{Q}_{S1}$ falls below 65% from Y2, triggering a £150/t scale-inefficiency penalty that inflates c_p to £2650/t.

Strategy 2 (Flexibility and Risk Mitigation). Invest £20M upfront in modular storage, demand-smoothing contracts, and digital forecasting. This does not expand capacity ($\bar{Q}$, s are unaffected), but instead reduces two key newsvendor cost drivers: demand volatility drops by 40% ($\sigma_t \to 0.6\sigma_t = 0.09\mu_t$), and pre-negotiated contingent contracts lower spot procurement cost to $c_s =$ £2900/t. The reduced c_s shifts the critical ratio from $F(Q^*) = 0.200$ to $F(Q^*) = 0.125$, pushing the optimal production quantity further below the mean; simultaneously, the tighter demand distribution ($\sigma \downarrow$) reduces both ELS and ELOI, cutting procurement and storage waste. Under high growth, capacity eventually binds ($Q_t^* > \bar{Q}$ from Y3), forcing systematic spot reliance.

Strategy 3 (Staged Expansion). Invest £8M upfront in expandable foundations and grid upgrades, preserving the option to scale later. This is a *real-options* structure: at the end of Y1, after observing demand growth, GreenCo chooses one of two branches:

- *High growth observed* $\rightarrow$ expand nominal capacity $K_{\text{nom}} = 65$ t/day at $\alpha = 95\%$ availability for £42M (discounted by the pre-installed foundations), matching S1 specifications from Y2 onward.
- *Low growth observed* $\rightarrow$ invest £15M in flexibility measures (a reduced version of S2: same σ reduction and $c_s =$ £2900/t), matching S2 specifications from Y2 onward.

The cost of deferral is a 10% realised revenue reduction in Y1–Y2 under both growth paths. Until the Y1 decision, capacity and parameters remain at baseline.

Table 3 summarises the parameter modifications. All unmodified parameters retain their base-case values from Table 1.

Table 3: Strategy-specific parameter modifications (base: $\bar{Q} = 1020$ t/mo, $s = 3$ bays, $c_s = £3200$/t, $\sigma_t = 0.15\mu_t$).

Parameter	Strategy 1	Strategy 2	Strategy 3
I_0 (Year 0)	£54M	£20M	£8M
I_1 (Year 1)	—	—	£42M (H) / £15M (L)
Capacity $\bar{Q}$	1852.5 t/mo	1020 t/mo	1020 → 1852.5 (H branch)
Availability	95%	85%	85% → 95% (H branch)
Bays s	5	3	3
σ_t	$0.15\mu_t$	$0.09\mu_t$	$0.15\mu_t$
c_s	£3200/t	£2900/t	£3200/t
Revenue	100%	100%	90% in Y1–Y2
Low-growth c_p penalty	+£150/t from Y2	—	—

3 Operational Risk Analysis

3.1 Monthly Production Decision: Newsvendor Model

Each month GreenCo chooses production quantity $Q \in [0, \bar{Q}]$ before demand $D \sim \mathcal{N}(\mu_t, \sigma_t^2)$ is realised. The effective monthly capacity is:

$$\bar{Q} = K_{\text{nom}} \cdot \alpha \cdot n_d \tag{2}$$

All demand is contractually met; shortfalls are covered by spot procurement. Monthly profit:

$$\Pi(Q, D) = \underbrace{pD}_{\text{revenue}} - \underbrace{c_p Q}_{\text{production}} - \underbrace{c_s \max(D - Q, 0)}_{\text{spot procurement}} - \underbrace{c_h \max(Q - D, 0)}_{\text{storage}} \tag{3}$$

Taking expectations over demand D, we define *expected lost sales* (demand unmet by own production) and *expected leftover inventory* (unsold overproduction):

$$\text{ELS}(Q) = \mathbb{E}_D[\max(D - Q, 0)], \qquad \text{ELOI}(Q) = \mathbb{E}_D[\max(Q - D, 0)] \tag{4}$$

Since revenue pD is independent of Q (contractual obligation), expected monthly profit is:

$$\mathbb{E}[\Pi(Q)] = p\mu_t - c_p Q - c_s\,\text{ELS}(Q) - c_h\,\text{ELOI}(Q) \tag{5}$$

3.1.1 Optimal Production Quantity

Note $\frac{d}{dQ}\text{ELS} = -(1 - F(Q))$ and $\frac{d}{dQ}\text{ELOI} = F(Q)$, where F is the CDF of demand D. Setting $d\mathbb{E}[\Pi]/dQ = 0$:

$$\frac{d\mathbb{E}[\Pi]}{dQ} = -c_p + c_s\big(1 - F(Q)\big) - c_h F(Q) = 0 \quad\Longrightarrow\quad \boxed{F(Q^*) = \frac{c_s - c_p}{c_s + c_h} = \frac{c_u}{c_u + c_o}} \tag{6}$$

This is the *newsvendor critical ratio*, where $c_u = c_s - c_p$ is the underage cost (extra cost of spot vs own production) and $c_o = c_p + c_h$ is the overage cost (production + storage on unsold units, with zero salvage). For normal demand $D \sim \mathcal{N}(\mu_t, \sigma_t^2)$, the capacity-constrained optimum at year t is:

$$Q_t^* = \min\big(\mu_t + z^*\sigma_t,\ \bar{Q}\big), \qquad z^* = \Phi^{-1}\left(\frac{c_u}{c_u + c_o}\right) \tag{7}$$

where Φ^{-1} is the inverse CDF of the standard normal distribution.

3.1.2 Expected Profit Components

For normal demand, the standard normal loss function $L(z) = \phi(z) - z[1 - \Phi(z)]$ gives closed-form expressions for ELS and ELOI. Writing $z_t = (Q_t^* - \mu_t)/\sigma_t$:

$$\text{ELS}(Q_t^*) = \sigma_t\, L(z_t) \qquad \text{(expected spot procurement volume)} \tag{8}$$

$$\text{ELOI}(Q_t^*) = \sigma_t\big[z_t + L(z_t)\big] \qquad \text{(expected unsold overproduction)} \tag{9}$$

The expected annual profit for strategy k in scenario j at year t is:

$$\mathbb{E}[\Pi_t^{(k,j)}] = 12\Big(\alpha_t^{(k)}\, p\mu_t^{(j)} - c_p^{(k)}\, Q_t^{*(k,j)} - c_s^{(k)}\, \mathrm{ELS}(Q_t^{*(k,j)}) - c_h\, \mathrm{ELOI}(Q_t^{*(k,j)})\Big) \tag{10}$$

where $\alpha_t^{(k)}$ is the revenue realisation factor ($\alpha = 0.9$ for S3 in Y1–Y2 due to deferral costs; $\alpha = 1$ otherwise), $\mu_t^{(j)}$, $\sigma_t^{(j)}$ come from Table 2, and $c_s^{(k)}$, $c_p^{(k)}$, $\bar{Q}^{(k)}$ are from Table 3.

3.1.3 Profit Evolution Under Various Demand Growth and Investment Strategies

When the capacity constraint does not bind, $z_t = z^*$ and Q_t^* scales linearly with μ_t. Once demand growth pushes $\mu_t + z^*\sigma_t > \bar{Q}$, the constraint binds: $Q_t^* = \bar{Q}$, z_t falls below z^*, and ELS grows rapidly as demand outpaces production. Under high growth with base capacity, this occurs from Year 2 onward. This is the principal mechanism forcing emergency spot procurement under Strategies 2 and 3 (before expansion).

Table 4 summarises expected annual profit $\mathbb{E}[\Pi_t^{(k,j)}]$ under each strategy $k \in \{S1, S2, S3\}$, scenario $j \in \{\textit{High-Growth+No Competition, High-Growth+Competition, Low-Growth}\}$, and year $t \in \{0, 1, 2, 3, 4\}$. We recalculate $\mathbb{E}[\Pi_t^{(k,j)}]$ using Eqn 10 based on the modelling changes made to our parameters: effective capacity $\bar{Q}$ under strategy k, σ_t, c_s and c_p etc.

Table 4: Expected annual profit $\mathbb{E}[\Pi_t^{(k,j)}]$ (£M) by strategy, scenario, and year.

Strategy	Scenario	Y1	Y2	Y3	Y4
S1	High, no comp.	17.86	19.65	21.61	23.77
	High + comp.	17.86	19.65	18.37	20.21
	Low growth	16.72	15.58	16.05	16.53
S2	High, no comp.	19.02	20.80	22.46	24.22
	High + comp.	19.02	20.80	19.56	21.29
	Low growth	17.81	18.34	18.89	19.46
S3	High, no comp.	12.58	13.84	21.61	23.77
	High + comp.	12.58	13.84	18.37	20.21
	Low growth	11.78	13.25	18.89	19.46

Key observations: (i) S2 consistently produces higher annual profit than S1 in every scenario and year, despite identical revenue—the reduced spot price (£2900/t) and tighter demand distribution (lower σ) jointly reduce procurement and storage costs; (ii) S1's low-growth profits fall sharply in Y2–Y4 due to the £150/t scale-inefficiency penalty; (iii) S3's Y1–Y2 profits are suppressed by the 10% revenue reduction, recovering only from Y3 once the Year-1 investment takes effect.

3.2 Refuelling Station Congestion: M/M/s Queue

HGVs arrive as a Poisson process at rate $\lambda = 10/\text{hr}$. Refuelling times are i.i.d. $\text{Exp}(\mu)$ with mean $1/\mu = 12$ min ($\mu = 5/\text{hr}$ per bay), and there are $s_0 = 3$ bays in parallel (FIFO, infinite buffer). This is an M/M/s queue with load $a = \lambda/\mu$ and server utilisation $\rho = \lambda/(s\mu)$. A steady state exists if and only if $\rho < 1$.

The idle probability, expected queue length, and queue waiting times are:

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{a^n}{n!} + \frac{a^s}{s!} \cdot \frac{1}{1-\rho}\right]^{-1} \tag{11}$$

$$L_q = \frac{a^s\, \rho}{s!(1-\rho)^2}\, P_0 \tag{12}$$

The remaining performance measures follow from Little's Law ($L = \lambda W$, $L_q = \lambda W_q$) and the decomposition $W = W_q + 1/\mu$, where W and L are the expected system waiting times and system size respectively:

$$W_q = \frac{L_q}{\lambda}, \qquad W = W_q + \frac{1}{\mu}, \qquad L = \lambda W \tag{13}$$

3.2.1 Baseline Performance ($s = 3$)

With $a = \lambda/\mu = 2$, $\rho = 2/3 < 1$ (stable), applying Eqns (11)–(13) gives $P_0 = 1/9$, $L_q = 8/9 \approx 0.889$ vehicles, and $W_q = L_q/\lambda =$ **5.33 min**. Even at baseline, $W_q = 5.33$ min $> W_q^{\max} = 5$ min. GreenCo is *already* non-compliant with its service level agreement (SLA) before any demand growth. The maximum arrival rate compatible with compliance is $\lambda^{\max}_{s=3} = 9.84$/hr, just below the current rate of 10/hr.

3.2.2 Impact of Hydrogen Demand Growth

Assuming $\lambda_t = \lambda_0(1+g)^t$, with HGV arrival rates scaling with mean hydrogen demand growth, Table 5 shows how W_q evolves. Under either growth scenario with $s = 3$, the wait time violates the 5-minute constraint from Year 0 and grows catastrophically—reaching 159.6 min by Y4 under high growth as $\rho \to 1$. With $s = 5$ bays (Strategy 1), $\rho = 0.40$ at baseline and W_q remains below 1.3 min across the entire 4-year horizon under any growth scenario. The maximum compliant λ for $s = 5$ is 19.13/hr, far exceeding any projected arrival rate.

Table 5: W_q (minutes) under demand growth. Values marked * exceed the 5-min target.

	High growth $g_H = 0.1$			**Low growth** $g_L = 0.03$		
Year	λ (/hr)	W_q, s=3	W_q, s=5	λ (/hr)	W_q, s=3	W_q, s=5
0	10.00	5.33*	0.24	10.00	5.33*	0.24
1	11.00	8.13*	0.36	10.30	6.04*	0.27
2	12.10	13.62*	0.54	10.61	6.87*	0.31
3	13.31	28.22*	0.82	10.93	7.88*	0.35
4	14.64	159.63*	1.27	11.26	9.10*	0.40

A fourth bay ($s = 4$) raises the compliant threshold to 14.44/hr, sufficient under low growth across all years but insufficient under high growth by Y4 ($\lambda_4 = 14.64$/hr). **Service-level compliance under any growth scenario requires adding at least two bays ($s = 5$), regardless of the chosen investment strategy.** This is a critical operational recommendation that Strategy 2 and 3 fail to address.

4 Strategic Investment Analysis

4.1 Decision Tree and NPV Formulation

The three strategies are evaluated over a 4-year horizon using a multi-stage decision tree (Figure 1). Strategy 3 has a real-options structure: at the end of Y1, GreenCo observes realised demand growth and selects a branch. If high growth is observed, automatically expand capacity (£42M, matching S1 from Y2); if low growth, invest in flexibility (£15M, matching S2 from Y2). The tree in Figure 1 shows this resolved optimal policy directly.

The Net Present Value (NPV) discounts future expected profits. The NPV for strategy k under scenario j is:

$$\text{NPV}^{(k,j)} = -I_0^{(k)} + \sum_{t=1}^{4} \frac{\mathbb{E}[\Pi_t^{(k,j)}]}{(1+r)^t} - \sum_{t\geq 1} \frac{I_t^{(k,j)}}{(1+r)^t} \tag{14}$$

where $I_0^{(k)}$ is the upfront capital expenditure (Table 3), $I_1^{(k,j)}$ is the conditional Year-1 investment discounted at $t = 1$ (applicable S3 only), and annual profits $\mathbb{E}[\Pi_t^{(k,j)}]$ are from Table 4.

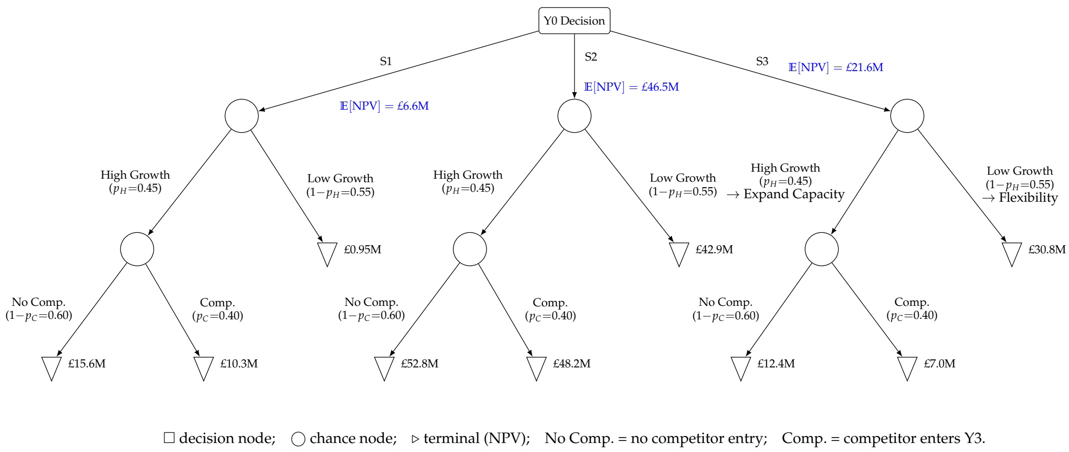

Figure 1: Decision tree for the three strategies. Terminal node values are NPVs. S3's Y1 decision is resolved by backward induction: if high growth is observed, expand capacity (matching S1 specifications); if low growth, invest in flexibility (matching S2 specifications). EMV values (blue) computed by backward induction.

4.2 NPV Results

Table 6: NPV by strategy and scenario (£M), with expected NPV.

Scenario	S1	S2	S3
High, no comp. ($p = 0.27$)	15.63	52.75	12.37
High + comp. ($p = 0.18$)	10.26	48.15	7.00
Low growth ($p = 0.55$)	0.95	42.93	30.83
E[NPV]	**6.59**	**46.52**	**21.55**

Strategy 2 dominates by a large margin in every scenario. Its E[NPV] of £46.5M exceeds S1 by £39.9M and S3 by £25.0M. The mechanism is threefold: (i) much lower capital expenditure (capex) (£20M vs £54M/£50M equivalent) leaves less invested capital to recover; (ii) reduced c_s from £3200 to £2900/t lowers the underage penalty, shifting the critical ratio from 0.200 to 0.125 and reducing expected spot procurement costs; (iii) reduced demand volatility ($\sigma \to 0.6\sigma$) lowers both expected leftover and expected shortfall, simultaneously reducing storage and procurement waste.

S1 performs poorly under low growth (NPV = £0.95M) due to the combination of high capex and the scale-inefficiency cost penalty. S3's real-options value—the ability to defer the large expansion cost—is insufficient to compensate for the 10% Y1–Y2 revenue penalty and the fact that Strategy 2 is already cheaper and more profitable in every state.

4.3 Monte Carlo Simulation Results

For each of $N = 50000$ simulated paths: (i) draw growth scenario and competitor entry; (ii) for each of 48 months draw $D_m \sim \mathcal{N}(\mu_m, \sigma_m^2)$ (clipped at zero); (iii) compute monthly profit using the capacity-constrained optimal Q_m^*; (iv) aggregate to annual totals and discount. Table 7 reports the full risk profile.

Table 7: Monte Carlo risk measures ($N = 50000$, all values in £M except $\mathbb{P}(\text{NPV} < 0)$). Conditional Value-at-Risk ($\text{CVaR}_{5\%}$): expected NPV in the worst 5% of outcomes.

Measure	S1	S2	S3
$\mathbb{E}[\text{NPV}]$	6.58	46.54	21.50
Std. deviation	6.71	4.35	10.51
$\text{CVaR}_{5\%}$	-1.97	41.19	5.01
$\mathbb{P}(\text{NPV} < 0)$	13.8%	0.0%	0.0%
10th percentile	-0.38	42.12	7.23
90th percentile	16.37	53.06	31.74

Each strategy's NPV distribution (Figure 2) is a mixture of three Gaussians, one per terminal scenario weighted by scenario probability (0.27, 0.18, 0.55). Within each scenario, monthly demand randomness produces a Gaussian-shaped NPV cluster; the separation between clusters reflects the scenario-driven differences in annual profits.

Strategy 2 is superior on *every* risk dimension simultaneously—it has the highest E[NPV], the *lowest* standard deviation, the highest CVaR, and zero probability of a net loss. Its three scenario clusters are tightly grouped in the £42–53M range (Figure 2), with little overlap—even the worst-case cluster (low growth) sits above £40M.

Strategy 1 has a 13.8% probability of a net loss. Its low-growth cluster (weight 0.55) is centred near £1M with left-tail mass crossing zero, while the two high-growth clusters sit around £10–16M. The wide spread (std. £6.71M) reflects large inter-scenario separation compounded by within-scenario demand variability.

Strategy 3 has zero probability of loss but a distinctly bimodal distribution (std. £10.51M): the dominant low-growth cluster (weight 0.55) sits near £31M, while the two high-growth clusters are compressed around £7–12M. The real option is more valuable under *low* growth—flexibility investment is cheap and the expensive capacity expansion is avoided.

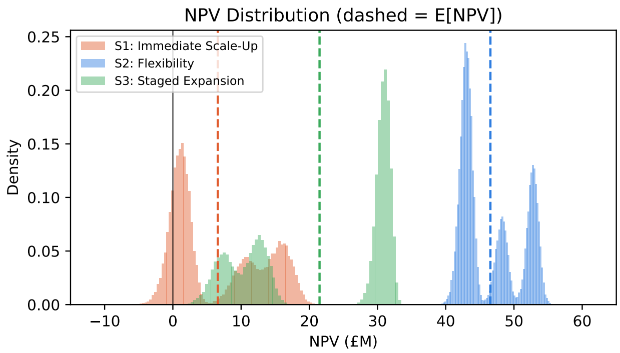

Figure 2: Simulated NPV density for each strategy (dashed lines = E[NPV]). Each distribution is a mixture of three Gaussian clusters corresponding to the three terminal scenarios. S2's clusters are tightly grouped above £40M; S1's low-growth cluster crosses zero; S3 is bimodal with the dominant low-growth peak near £31M.

5 Conclusions

Our analysis aims to maximise expected NPV while explicitly assessing downside risk exposure and service-level compliance across GreenCo's four-year investment horizon.

5.1 Operational Risks

Four risk sources were analysed: (i) demand volatility, addressed via the newsvendor model and Monte Carlo simulation; (ii) production capacity constraints, quantified through the capacity-constrained newsvendor under each strategy; (iii) congestion risk, evaluated with the $M/M/s$ queueing model; and (iv) long-term investment uncertainty (demand trajectory, competitor entry), structured through the decision tree and scenario analysis. The principal finding is that under GreenCo's cost structure ($c_o \gg c_u$), demand uncertainty—not capacity shortfall—is the dominant driver of profit variability.

5.2 Recommendations

The following recommendations enhance resilience, improve performance stability, and support informed long-term investment decisions under uncertainty:

1. **Adopt Strategy 2 (long-term investment decision).** S2 achieves the highest E[NPV] of £46.5M (vs £6.6M for S1, £21.6M for S3), zero probability of net loss, lowest NPV volatility, and highest CVaR. It dominates in every scenario (Table 6). The mechanism: lower capex (£20M), reduced spot cost ($c_s = £2900/\text{t}$ shifting the critical ratio from 0.200 to 0.125), and reduced demand volatility ($\sigma \to 0.6\sigma$) jointly cut procurement and storage waste. Under high growth, S2 relies on spot procurement from Y3, but pre-negotiated contracts at £2900/t provide downside protection; even in this worst case, S2's NPV (£48–53M) far exceeds alternatives.

2. **Implement newsvendor-optimal production (performance stability).** Set $Q_t^* = \min(\mu_t + z^*\sigma_t, \bar{Q})$, reviewed annually. Planning at mean demand ignores the asymmetric cost structure ($c_o \gg c_u$) and the flaw of averages, leading to systematic overproduction and unnecessary cost exposure.
3. **Add two refuelling bays immediately (service-level compliance).** The current M/M/3 configuration violates the 5-minute SLA at baseline ($W_q = 5.33$ min) and deteriorates rapidly under demand growth. Two additional bays ($s = 5$) are required for compliance across the full horizon ($W_q \leq 1.27$ min). A single bay ($s = 4$) is insufficient under high growth by Y4. This is independent of the investment strategy chosen.
4. **Monitor demand and maintain adaptive capacity (resilience).** If high growth materialises earlier than expected, activating an option to expand capacity (analogous to S3) may become worthwhile from Y2. GreenCo should also secure spot contract volumes sufficient to cover projected shortfalls under high growth and establish fallback suppliers to mitigate counterparty risk.

5.3 Assumptions and Limitations

- **S3 specification matching:** The coursework does not specify availability or demand parameters for S3's conditional investments. We assume S3's expansion branch matches S1 specifications (95% availability, 65 t/day) and S3's flexibility branch matches S2 specifications ($\sigma \to 0.6\sigma$, $c_s = £2900$/t).
- **Arrival rate scaling:** HGV arrival rate λ_t is assumed to scale with mean hydrogen demand growth. This links the queueing and newsvendor models but is not specified in the problem.
- **Demand independence:** Monthly demands are assumed i.i.d. within each year. Serial correlation would alter the newsvendor solution and widen the simulated NPV distribution.
- **Risk neutrality:** The analysis maximises E[NPV]. A risk-averse decision-maker would weight downside outcomes more heavily, strengthening the case for S2 further.
- **Queueing model:** Steady-state M/M/s assumes stationary Poisson arrivals. Time-of-day arrival patterns would give a more accurate picture of peak congestion.